

Program 10

Demonstrate Kubernetes dashboard & CLI for Cluster Monitoring

Steps:

1. Create 2 files: deployment.yaml and service.yaml

Inside deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Inside service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
  type: NodePort
```

2. Apply both .yaml files.

```
(1rv24mc045_ryan@ ryan-kali) [~/Documents/program10]
$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created

(1rv24mc045_ryan@ ryan-kali) [~/Documents/program10]
$ kubectl apply -f service.yaml
service/nginx-service created
```

3. Use CLI to list out all pod and services

```
(1rv24mc045_ryan@ ryan-kali) [~/Documents/program10]
$ kubectl get pods
NAME                                READY STATUS RESTARTS AGE
nginx-deployment-6f9664446b-bhhrp  1/1   Running  0      17s
nginx-deployment-6f9664446b-l7xd5  1/1   Running  0      17s

(1rv24mc045_ryan@ ryan-kali) [~/Documents/program10]
$ kubectl get svc
NAME          TYPE        CLUSTER-IP   EXTERNAL-IP  PORT(S)    AGE
kubernetes    ClusterIP   10.96.0.1    <none>       443/TCP    6d23h
nginx-service  NodePort    10.110.76.171 <none>       80:32463/TCP 12s
```

4. Open Kubernetes dashboard to observe the deployments

```
(1rv24mc045_ryan@ ryan-kali) [~/Documents/program10]
$ minikube dashboard
Enabling dashboard ...
```

The screenshot shows the Kubernetes dashboard interface. The top navigation bar includes the 'kubernetes' logo, a dropdown menu set to 'default', and a search bar. The main content area is titled 'Workloads' and features three large green circular icons representing 'Deployments', 'Pods', and 'Replica Sets'. Below these icons, there are three detailed views:

- Deployments:** A table with columns 'Name', 'Images', 'Labels', 'Pods', and 'Created'. It shows one entry: 'nginx-deployment' with image 'nginx:latest', label 'app: nginx', 2/2 pods, and created 16 minutes ago.
- Pods:** A table with columns 'Name', 'Images', 'Labels', 'Node', 'Status', 'Restarts', 'CPU Usage (cores)', 'Memory Usage (bytes)', and 'Created'. It shows two entries for 'nginx-deployment-6f9664446b', both in 'Running' status on 'minikube' node.
- Replica Sets:** A table with columns 'Name', 'Images', 'Labels', 'Pods', and 'Created'. It shows one entry: 'nginx-deployment-6f9664446b' with image 'nginx:latest', label 'app: nginx', 2/2 pods, and created 16 minutes ago.

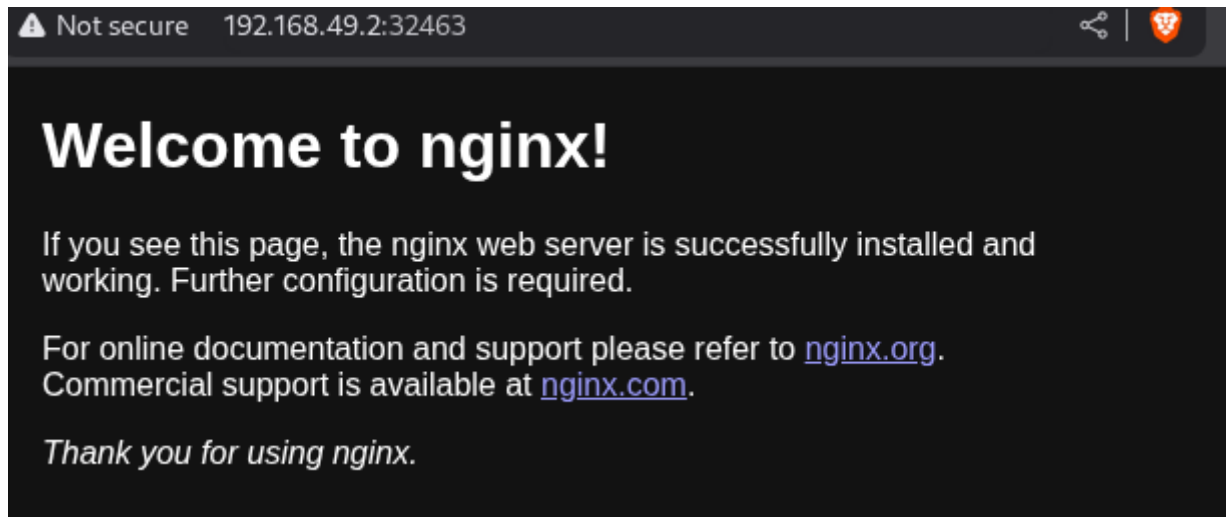
The left sidebar contains a list of navigation links: Workloads, Cron Jobs, Daemon Sets, Deployments, Jobs, Pods, Replica Sets, Replication Controllers, Stateful Sets, Service, Ingresses, Ingress Classes, Services, Config and Storage, Config Maps, Persistent Volume Claims, Secrets, Storage Classes, Cluster, Cluster Role Bindings, Cluster Roles, Events, Namespaces, and Network Policies.

5. Open the deployment in the browser.

```
(1rv24mc045_ryan@ryan-kali) ~/Documents/program10
$ minikube service nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:32463

Opening service default/nginx-service in default browser...



6. Delete all pods and services.

```
(1rv24mc045_ryan@ryan-kali) ~/Documents/program10
$ kubectl delete deployments --all
deployment.apps "nginx-deployment" deleted
```